

MIP_ARCHITECTURE.md

Mainframe Intelligence Platform

Technical Architecture

Version: 1.0

Architectural Principles

Principle 1

Facts Before AI

Source Systems ↓ Metadata ↓ Knowledge Graph ↓ Reasoning ↓ AI

AI must consume verified enterprise knowledge.

AI must not become the source of truth.

Principle 2

Technology Agnostic

Support:

- COBOL
- PL/I
- JCL
- DB2
- VSAM
- IMS
- CICS
- MQ
- Control-M
- CA7

Future support:

- Java
 - .NET
 - Python
 - Microservices
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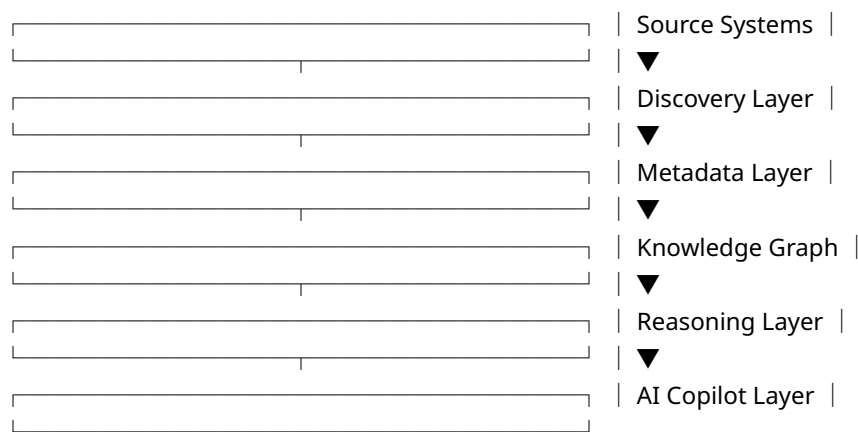
Principle 3

Incremental Discovery

The platform must continuously ingest and update metadata.

No full re-scan requirement.

High-Level Architecture



Discovery Layer

Purpose:

Extract metadata from source artifacts.

Components:

Repository Scanner

Responsibilities:

- Repository crawling
- File classification
- Change detection

COBOL Parser

Extract:

- PROGRAM-ID
 - CALL statements
 - COPY statements
 - EXEC SQL
 - File definitions
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JCL Parser

Extract:

- JOB
 - PROC
 - EXEC PGM
 - DD statements
-

DB2 Parser

Extract:

- Tables
 - Views
 - Indexes
 - Constraints
-

Copybook Parser

Extract:

- Structures
 - Fields
 - Data Types
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Scheduler Parser

Extract:

- Job dependencies
 - Calendars
 - Execution chains
-

Metadata Layer

Storage:

PostgreSQL

Purpose:

Store normalized metadata.

Core Tables:

- programs
 - jobs
 - copybooks
 - tables
 - datasets
 - transactions
 - business_capabilities
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Knowledge Graph Layer

Storage:

Neo4j

Purpose:

Store relationships.

Nodes:

- Program
- Job
- Table
- Copybook
- Dataset
- Transaction
- Business Capability
- Business Rule

Relationships:

- CALLS
- EXECUTES
- READS
- WRITES

- USES
 - UPDATES
 - IMPLEMENTS
 - BELONGS_TO
-

Reasoning Layer

Purpose:

Enterprise intelligence.

Capabilities:

Root Discovery

Find entry points.

Flow Discovery

Trace execution paths.

Dependency Analysis

Find downstream dependencies.

Impact Analysis

Determine change impact.

Business Rule Discovery

Extract and explain rules.

AI Copilot Layer

Purpose:

Natural language interaction.

Questions:

- Which jobs execute program X?
- Show payment posting flow.
- Which programs update table Y?
- Explain this COBOL module.

Architecture:

User ↓ Copilot ↓ Reasoning Layer ↓ Knowledge Graph ↓ Metadata

Recommended Technology Stack

Backend:

- Python 3.13
- FastAPI
- Pydantic v2

Metadata Store:

- PostgreSQL

Knowledge Graph:

- Neo4j

Graph Processing:

- NetworkX

Caching:

- Redis

Messaging:

- Kafka

Vector Search:

- OpenSearch

AI Framework:

- LangGraph

Observability:

- OpenTelemetry
- Prometheus

- Grafana

Containerization:

- Docker
- Kubernetes

CI/CD:

- GitHub Actions
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Scalability Requirements

Support:

- Enterprise repositories
 - Continuous ingestion
 - Incremental updates
 - Multi-team access
 - Multi-application support
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Security

Requirements:

- RBAC
 - Audit Trails
 - Encryption
 - Data Classification
 - Source Code Governance
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Future Extensions

- COBOL to Java
 - Service Candidate Generation
 - API Discovery
 - Event Discovery
 - Automated Test Generation
 - Modernization Roadmaps
 - Enterprise Architecture Integration
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End State

MIP becomes the system of intelligence for legacy applications and modernization programs.